

Named Entity Recognition (NER) Guidelines for Moroccan Dialect (Darija)

I. Introduction

This document provides guidelines for annotating Named Entities (NEs) in the Moroccan dialect (Darija). The goal is to create a consistent annotation scheme for identifying and tagging entities in unstructured texts, facilitating the development of Natural Language Processing (NLP) systems for Darija.

These guidelines draw inspiration from Sang and De Meulder (2003) with adaptations to account for linguistic and cultural features specific to Moroccan Arabic.

II. General Principles

1. Annotators must tag all Named Entities (NEs) explicitly mentioned in the text.
2. Each NE is tagged with its full extent and assigned a semantic type.
3. Only entities that are proper names or explicit references to identifiable entities are annotated.
4. Ambiguities are resolved by considering the entity's function and context within the text.

The following four main entity types are considered:

- **Person (PER)**
- **Organization (ORG)**
- **Location (LOC)**
- **Other (OTH)**

1. Types of Named Entities

1.1. Person Entities (PER)

Person Entities include the names of individuals, groups of people, nicknames, and titles when directly associated with names.

Annotation Examples:

- [محمد]B-PER (Mohamed)
- [عبد الله]B-PER [كنون]I-PER (Abdullah Kanoun)
- [الملك]B-PER [محمد]I-PER [السادس]I-PER (King Mohammed VI)
- [نوال]B-PER [المطوكل]I-PER (Nawal El Moutawakel)

Rules:

1. Titles preceding a name (e.g., "الشيخ" or "الملك") are included in the annotation.
2. Nicknames or descriptive terms used to refer to people are also annotated:
 - Example: [السبع]B-PER (Lion, a nickname).
3. When a person's name appears in possessive constructions, only the name is annotated:
 - Example: [فاطمة] الكتاب B-PER (Fatima's book).

1.2.Organization Entities (ORG)

Organization Entities include the names of formally established groups, institutions, and companies.

Annotation Examples:

- [حكومة]B-ORG [المغرب]I-ORG (Government of Morocco)
- [البنك الشعبي]B-ORG (Banque Populaire)
- [شركة]B-ORG [اتصالات المغرب]I-ORG (Maroc Telecom)
- [فريق]B-ORG [الوداد]I-ORG (Wydad Team)

Rules:

1. When an organization is referred to by its acronym, the acronym is annotated:
 - Example: [لارام]B-ORG (RAM).
2. Sub-units within organizations are annotated separately:
 - Example: [وزارة]B-ORG [التعليم]I-ORG (Ministry of Education).
3. Names of political parties, NGOs, and media outlets are annotated as ORG:
 - Example: [العدالة والتنمية]B-ORG (Justice and Development Party).

1.3.Location Entities (LOC)

Location Entities include geographical locations, political regions, and landmarks.

Annotation Examples:

- [المغرب]B-LOC (Morocco)
- [تطوان]B-LOC (Tetouan)

- [بلاسا]B-LOC [جامع د الامة]I-LOC (Plasa Mosque of the Nation)

- [جبال]B-LOC [الريف]I-LOC (Rif Mountains)

Rules:

1. Locations in compound phrases are annotated fully:

- Example: [مدينة]B-LOC [طنجة]I-LOC (City of Tangier).

2. Addresses are annotated to include street names and numbers:

- Example: [شارع]B-LOC [محمد الخامس]I-LOC (Mohammed V Avenue).

3. Natural landmarks (mountains, rivers) are annotated as LOC:

- Example: [وادي]B-LOC [اللوكوس]I-LOC (Oued Loukkos).

1.4.Other Entities (OTH)

The "OTHER" category includes entities that do not fall under Person, Organization, or Location but are still significant. This may include products, events, artworks, awards, or other named entities.

Annotation Examples:

- [جائزة المغرب للكتاب]B-OTH (Morocco Book Award)

- [العيد الكبير]B-OTH (Eid Al-Kbir)

- [مهرجان حب الملوك]B-OTH (Cherry Festival)

- [الآيفون]B-OTH (iPhone)

Rules:

1. Products, such as brands or devices, are annotated:

- Example: [سامسونج]B-OTH (Samsung).

2. Events and holidays are annotated:

- Example: [رأس السنة]B-OTH (New Year).

3. Titles of artworks, books, or films are annotated:

- Example: [كتاب]B-OTH [رواء مكة]I-OTH (Rawaa Makkah).

2. Special Cases

2.1.Ambiguous Entities

Entities that could belong to more than one category (e.g., metonymic uses) are annotated based on their function in context.

Examples:

1. [الرباط]B-LOC as a city vs. [الرباط]B-ORG as the government.
2. [وداد]B-ORG as a football team vs. [وداد]B-PER as a person's name.

2.2.Non-Named Entities

Certain terms and expressions are not annotated as Named Entities, even if they resemble proper names.

Examples:

1. Titles of books or songs are not annotated unless they explicitly contain entities:
 - Example: "أغنية قمر" (Song "Qamar") is not annotated.
2. Names used in idiomatic expressions are not annotated:
 - Example: [تازة] ما بعيدة غي (Not far from Taza, an idiomatic expression to reduce the distance traveled).

3. Annotation Format

Entities are annotated using the **BIO** (Begin-Inside-Outside) tagging scheme:

- **B-TYPE**: Marks the beginning of an entity.
- **I-TYPE**: Marks the continuation of the same entity.
- **O**: Marks words that are not part of an entity.

Example Sentence:

[المغرب] B-LOC [نظم ماتش بين [الرجاء] B-ORG [البيضاوي] B-ORG و[افسي برشلونة] I-ORG [استاد] B-LOC [بن بطوطة] I-LOC
[ولي العهد] B-PER [الأمير] I-PER [الحسن] I-PER.

Translation:

(Morocco organized a match between Raja Casablanca and FC Barcelona at Ibn Battuta Stadium, attended by Crown Prince Hassan.)

Conclusion

This guideline provides a framework for consistently annotating Named Entities in Moroccan Arabic (Darija). It balances linguistic nuances with practical annotation practices to support NLP applications.